

Advanced (Carolina Reaper)

- Q1.** Factor out the greatest common factor from $2x + 4$
(A) $2(x + 2)$
(B) $4(x + 1)$
(C) $2(x - 2)$
(D) $x(2 + 4)$
(E) $x(4 + 2)$
- Q2.** Simplify the expression $3x^2 + 6x$ by factoring
(A) $3x(x + 2)$
(B) $3x(x - 2)$
(C) $6x(x + 1)$
(D) $6x(x - 1)$
(E) $9x(x + 1)$
- Q3.** What is the greatest common factor of $12x$ and $8y$?
(A) 2
(B) 4
(C) 8
(D) 12
(E) 16
- Q4.** Factor out the greatest common factor from $9x^2 + 12x$
(A) $3x(3x + 4)$
(B) $3x(3x - 4)$
(C) $9x(x + 1)$
(D) $9x(x - 1)$
(E) $12x(x + 1)$
- Q5.** Simplify the expression $x^2 + 2x$ by factoring
(A) $x(x + 1)$
(B) $x(x - 1)$
(C) $x(x + 2)$
(D) $x^2(1 + 2)$
(E) $x^2(1 - 2)$
- Q6.** What is the greatest common factor of $15x$ and $20y$?
(A) 1
(B) 5
(C) 10
(D) 15
(E) 20
- Q7.** Factor out the greatest common factor from $6x^2 + 12x$
(A) $6x(x + 2)$
(B) $6x(x - 2)$
(C) $12x(x + 1)$
(D) $12x(x - 1)$
(E) $3x(2x + 4)$
- Q8.** Simplify the expression $8x + 16$ by factoring
(A) $8(x + 2)$
(B) $8(x - 2)$
(C) $16(x + 1)$
(D) $16(x - 1)$
(E) $4(2x + 4)$

Open Ended Questions (FRQ)

Q9. Factor out the greatest common factor from $4x^3 + 12x^2 + 8x$ and simplify the expression

Q10. Simplify the expression $9x^2 + 12x + 3$ by factoring, if possible

Chef's Tips

- Chef's Tips: Emphasize the importance of factoring out the greatest common factor
- Chef's Tips: Use visual aids to illustrate the concept of factoring
- Chef's Tips: Encourage students to simplify expressions using factoring